

System Architecture and Workflow Overview

System Architecture – Entity Relationship Diagram (ERD)

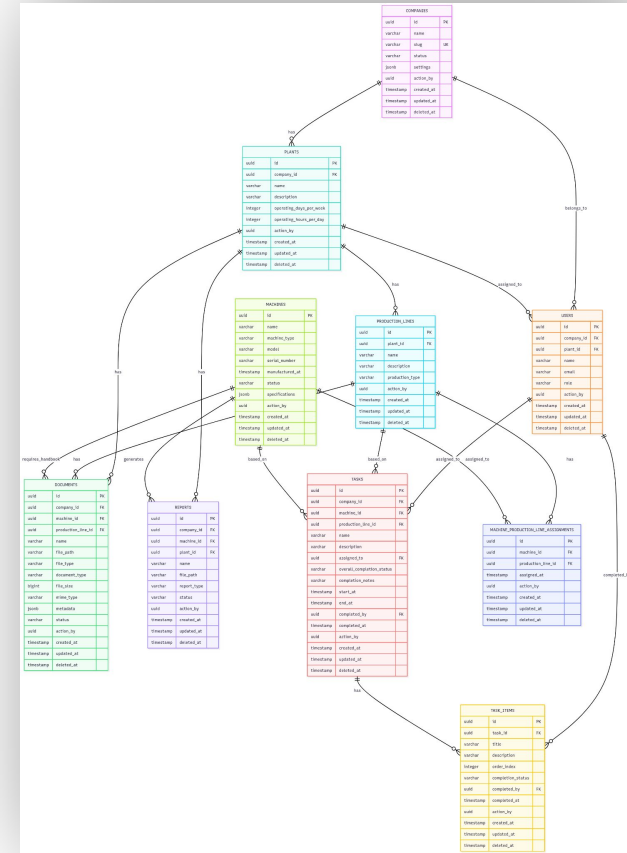


This system is organized around a hierarchical structure of plants, production lines, and machines.

Key components include:

- **User & Access Control:** Users are assigned roles and scoped to specific plants.
- **Asset Management:** Each plant contains production lines, which contain machines.
- **Document & Task Management:** Documents, tasks, and reports are linked to machines for support traceability and operations.

The architecture ensures every uploaded document, task, and user action is associated with a specific machine and plant.



User Creation and Access Management

1. **Laiq Team provides login credentials**

The Laiq team issues a company manager (Admin) login to the organization.

2. **Company Manager logs in**

The Admin accesses the Laiq system using the provided credentials.

3. **Manager creates Plants**

The Admin sets up one or more Plants representing operational sites or divisions.

4. **Manager selects a Plant and creates a User**

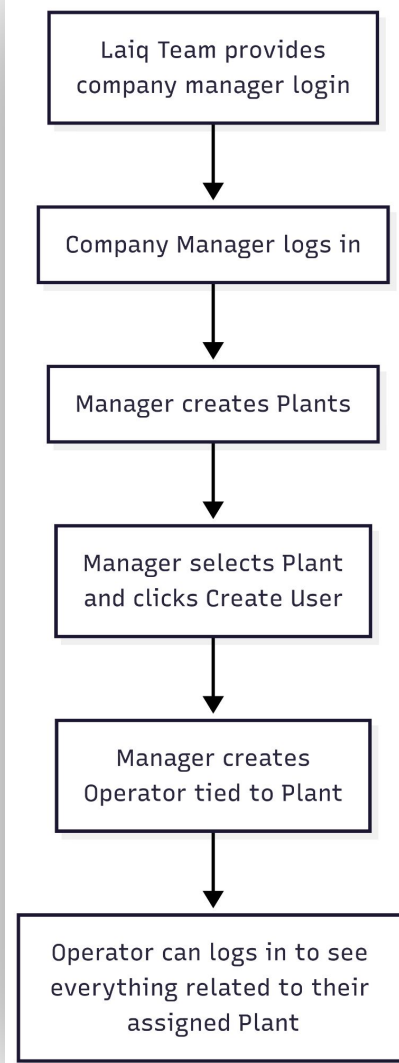
Within a selected Plant, the Admin initiates the user creation process.

5. **Manager creates an Operator tied to the Plant**

The new user is assigned the Operator role and is linked to the selected Plant.

6. **Operator logs in to access Plant-specific information**

The Operator can now log in and view all data and functions related to their assigned Plant only.

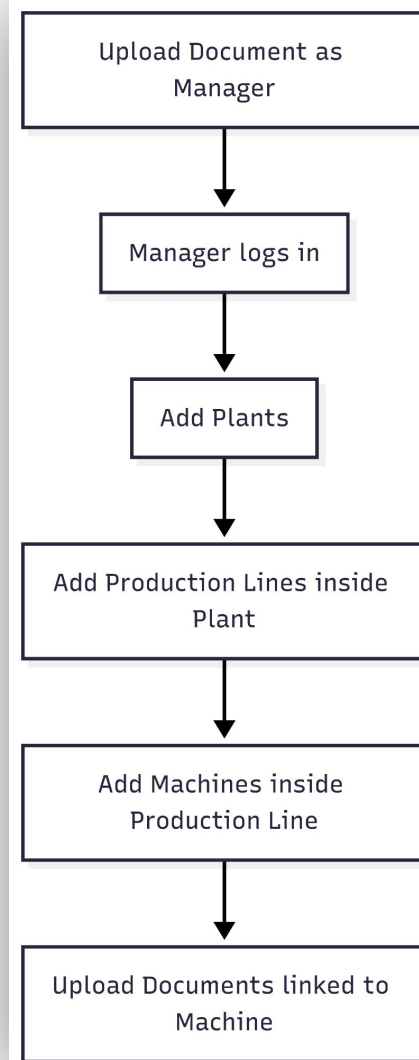


Document Upload Process for Managers

The document management flow is initiated by Managers:

1. Manager logs into the system.
2. Manager creates new plants.
3. Within each plant, the manager defines production lines.
4. Machines are added inside each production line.
5. Documents are uploaded and linked to specific machines.

This structure ensures documents are always associated with the correct machine and plant context, improving traceability, searchability, and maintenance operations.

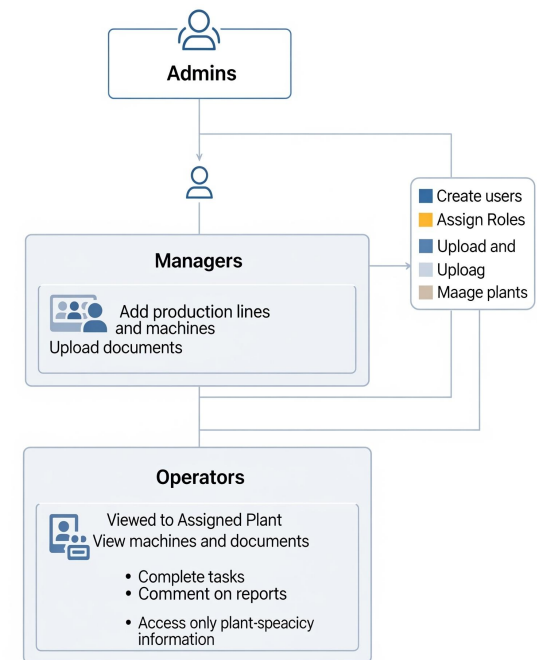


Role-Based Access Control

Each user in the system is assigned a specific role with clear permissions. Access is scoped to individual plants.

Role	Key Permissions
Admin	Create users, assign roles, manage plants
Manager	Add production lines and machines, upload and view documents
Operator	View machines and documents, complete tasks, comment on reports

User Roles and Permissions



1. Machine List

- Company 1
- Maintenance System
- Navigation
- Machines
- Operators
- Technicians
- Documents
- Reports
- Query

Dashboard

Total Machines: 25

Active Alerts: 9

Showing 10 of 25 machines

☐ Select page

Select all filtered

Status: All

Add Machine

Machine IPM-001

Working

Production Line: Assembly Line A

Factory: North Plant

Last Maintenance: 2024-01-15

Consult with AI

Machine IPM-002

Fixing

Production Line: Assembly Line A

Factory: North Plant

Last Maintenance: 2023-12-28

Consult with AI

Machine IPM-003

Working

Production Line: Quality Control Line

Factory: South Plant

Last Maintenance: 2024-01-08

Consult with AI

Machine IPM-004

Needs Fix

Production Line: Packaging Line B

Factory: South Plant

Last Maintenance: 2023-11-20

Consult with AI

Machine IPM-005

Working

Production Line: Storage Line A

Factory: East Plant

Last Maintenance: 2024-01-22

Consult with AI

Machine IPM-006

Fixing

Production Line: Loading Line

Factory: East Plant

Last Maintenance: 2023-12-15

Consult with AI

Machine IPM-007

Working

Production Line: Assembly Line B

Factory: North Plant

Last Maintenance: 2024-01-10

Consult with AI

Machine IPM-008

Fixing

Production Line: Testing Line

Factory: West Plant

Last Maintenance: 2023-12-05

Consult with AI

Machine IPM-009

Working

Production Line: Assembly Line A

Factory: North Plant

Last Maintenance: 2024-01-15

Consult with AI

Machine IPM-010

Fixing

Production Line: Assembly Line A

Factory: North Plant

Last Maintenance: 2023-12-28

Consult with AI

2. Add machines

Company 1

Maintenance System

Navigation

Machines

Operators

Technicians

Documents

Reports

Query

Dashboard

Total Machines: 25

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Select page

Select all filtered

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+ Add Machine

Machine IPM-001

Working

Production Line: Assembly Line A

Factory: North Plant

Last Maintenance: 2024-01-15

Consult with AI

Machine IPM-003

Working

Production Line: Quality Control Line

Factory: North Plant

Last Maintenance: 2024-01-08

Consult with AI

Machine IPM-004

Needs Fix

Production Line: Packaging Line B

Factory: South Plant

Last Maintenance: 2023-11-20

Consult with AI

Machine IPM-005

Working

Production Line: Storage Line A

Factory: East Plant

Last Maintenance: 2024-01-22

Consult with AI

Machine IPM-007

Working

Production Line: Assembly Line B

Factory: North Plant

Last Maintenance: 2024-01-10

Consult with AI

Machine IPM-008

Fixing

Production Line: Testing Line

Factory: West Plant

Last Maintenance: 2023-12-05

Consult with AI

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Add New Machine

Enter the machine details below. The machine will be added with "starting" status.

Machine Name *

Enter machine name

Production Line *

Enter production line name

Factory *

Enter factory name

Upload Documents

Choose Files

No file chosen

⬆

Cancel

Add Machine

Showing 2 of 2 machines ☒ Select page

Deselect all filtered

🔍 Status: Needs fix ▾

+ Add Machine

2 machines selected

Clear Selection

Actions: **Change Status** ▾

Machine IPM-004



Needs Fix

Production Line: Packaging Line B

 Factory: South Plant

 Last Maintenance: 2023-11-20

 Consult with AI

Machine IPM-010



 Needs Fix

Production Line: A

 Factory: South Plant

 Last Maintenance: 2023-11-08

 Consult with AI

4. Consult with AI



Company 1

Maintenance System



Navigation

 Machines

 Operators

 Technicians

 Documents

 Reports

 Consult

Machine Context

Machine IPM-001

 Working

 Production Line: Assembly Line A

 Factory: North Plant

 Last Maintenance: 2024-01-15

AI Assistant - Machine IPM-001



Hello! I'm here to help you with Machine IPM-001. I can see it's currently working and located on Assembly Line A at North Plant. What would you like to know or discuss about this machine?

7:38:52 PM

5. Operator List

 **Company 1**
Maintenance System

Navigation

 Machines

 Operators

 Technicians

 Documents

 Reports

 Consult



Operators

Manage equipment operators and assignments

[+ Add Operator](#)



Sarah Chen

Senior Operator

OP-001

Assigned Machine

IPM-001
Assembly Line A

working

 Edit



Michael Rodriguez

Mid Operator

OP-002

Assigned Machine

IPM-003
Quality Control Line

working

 Edit



Emily Johnson

Junior Operator

OP-003

Assigned Machine

IPM-002
Assembly Line A

fixing

 Edit



David Park

Senior Operator

OP-004

Assigned Machine

IPM-005
Storage Line A

working

 Edit



Technicians

Manage technical specialists assigned to critical machine issues

Technicians Management

+ Add Technician

Alex Rodriguez
TECH-001

Senior Maintenance Engineer

Assigned Machine:
Packaging Line B
IPM-004

needs fix

 Edit Delete

Jennifer Kim
TECH-002

 **Electrical Systems Specialist**

Assigned Machine:
Loading Line
IPM-006
fixing

 Edit

Delete

Marcus Thompson
TECH-003

Hydraulic Systems Technician

Assigned Machine:
Testing Line
IPM-008
fixing

 Edit Delete

Sofia Martinez
TECH-004

Ryan Chen
TECH-005

Amanda Davis
TECH-006